

Lite Foil remote

Datasheet & user manual

Firmware	1.15.0 (October 17, 2026)
Screen	1.69" color touchscreen, 240 × 280
Compatibility	Lite Foil eFoil, optional GPS/4G tracker
Waterproofing	IP69K
Weight	245 g
Battery	5,750 mAh
App	Lite Foil Android iOS
Languages	FR · EN · ES · DE



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Introduction

The Lite Foil remote controls the board by radio, shows you live everything the rider needs to know and runs the board's services: online GPS tracking, emergency alert, camera, timer. You hold it in one hand, trigger under your index finger, screen right in front of your eyes, and everything is done with two buttons or with a finger on the screen.



Top: screen (1), button A on the left, button B on the right



Side: trigger (2) under the index finger



Grip: charging port (3) at the bottom

Orientation: remote in hand, screen facing you, the blue pad under the screen carries two buttons. **A is the left one** (power symbol: a circle broken by a line), **B is the right one** (lightning symbol ϟ). The rest of the manual uses these two letters throughout.

1 · 1.69" touchscreen: color, 240 × 280, readable in full sunlight. Touch works only in the menus, never while riding or with the motor armed.

A · Left button (power): power on (3 s), arm / disarm the motor, confirm, go back, power off (2 s).

B · Right button (lightning): power mode, timer (3 presses), emergency (5 s), “next” and “change” in the menus.

A + B together (150 ms): opens and closes the menu from the riding screen.

2 · Trigger: contactless magnetic sensor (Hall effect), no wearing parts, read 50 times per second, guided calibration at every power-on.

3 · Charging port at the bottom of the grip: plugging in disarms and locks riding, and turns on Bluetooth for the app.

Inertial unit: 6-axis accelerometer and gyroscope for the tilt menus, falls, flights and heading (section 16).

Radio: dedicated 2.4 GHz link with the board’s receiver, protected pairing, motor cut-off within 0.5 s if the link drops.

02 Technical characteristics

Hardware

Processor	ESP32-S3 dual-core, 240 MHz, 16 MB flash memory (two firmware slots for over-the-air updates)
Screen	1.69" IPS color 240 × 280, 65,000 colors, LED backlight; all text in a rounded font (Nunito, 8 built-in sizes)
Touch	Capacitive (CST816), software detection of taps and swipes
Trigger	Hall-effect sensor, 16 averaged readings, adaptive filter (smooth at rest, responsive when moving)
Buttons	2 waterproof buttons A and B, short / long / combined presses
Inertial unit	QMI8658 6-axis (accelerometer + gyroscope): tilt menus, fall detection, smoothed heading
Clock	PCF85063 real-time clock: time available from power-on, set by the app, corrected by GPS
Logbook	3 MB of flash memory: every session recorded second by second, read back in the app
Waterproofing	IP69K (immersion and high-pressure jets)
Weight	245 g
Battery	1-cell Li-ion 5,750 mAh, managed by a dedicated BMS, % gauge on screen
Charging	Dedicated port at the bottom of the grip; “CHARGING” screen with % and voltage; automatic shutdown 2 min after full charge
Standby	Deep sleep: wake up with a long press on A (3 s) or by the charger

Links

Board	2.4 GHz radio (ESP-NOW), 50 commands/s to the receiver, telemetry 10 times/s in return, maximum transmit power
Pairing	By proximity (less than one meter), 5 s window when the board powers on, only one remote linked, address filtering
Bluetooth	Bluetooth Low Energy: Lite Foil app (settings, gauges, update) and GoPro / Insta360 cameras
Update	Wireless from the app, remote on the charger; settings and pairing are kept
GPS	In the board (receiver): speed, time, satellites sent up to the remote
Tracker (option)	5 Hz GPS + 4G modem in the board: live tracking on sportstracklive.com, emergency alert

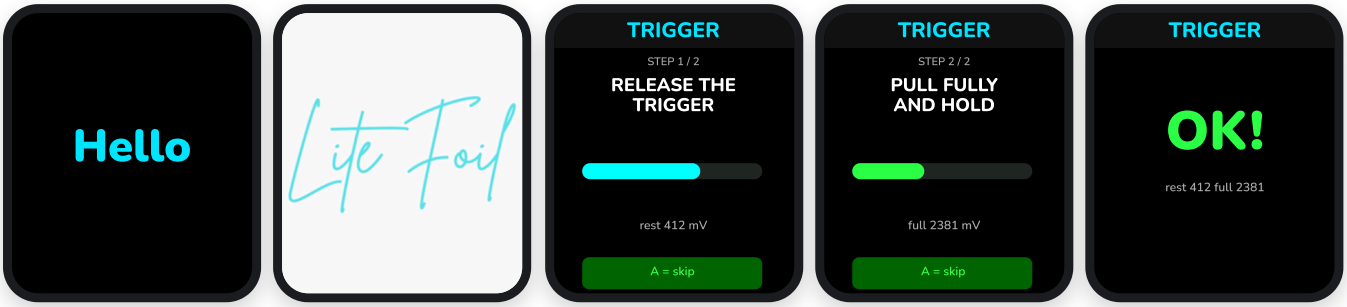
What it shows and controls

Function	Detail
Riding	GPS speed, throttle arc, LOW / MED / HIGH power mode, top speed of the session, **estimated range** (minutes and km remaining), adjustable **GPS speed limit**, board battery in %, voltage, power in watts, controller temperature, radio signal quality, satellites, time
Safety	Red “DISARMED” screen as long as the motor is locked, deliberate arming with the trigger at rest, adjustable auto disarm, cut-off within 0.5 s without a link, no power-off while the motor is armed
Riding settings	3 power modes (60 / 80 / 100 %), 4 sensitivity curves, startup boost 0 to 24 %, auto disarm 1 to 10 s, guided trigger calibration at every power-on
Services	SportsTrackLive live tracking, emergency alert to your loved ones, timer with history (time, V.MAX, distance), GoPro / Insta360 camera control (REC, mode, auto-REC)
Customization	km/h or mph, 2 screen themes (large speed or large battery), 4 languages, number of cells in the pack (8 to 20 S), radio ping display, lifetime odometer
Comfort	“Carousel” menu of 17 colored tiles, finger swipe or buttons, smartphone-style toggle pages, automatic standby after 20 min without any action

03 Getting started

Power on

Hold A for **3 seconds**. A shorter press won't wake the remote, so it can't switch on by accident in a bag. The sequence takes about 5 seconds: a big “Hello” in the chosen language, then the Lite Foil logo, then the guided trigger calibration, then the red “DISARMED” screen. **The remote always starts disarmed.**



1. Hello
2 seconds

2. Logo
3 seconds

3. Trigger, step 1
release; the bar fills in 1.5 s

3b. Step 2
only if the rest position moved: pull fully

4. OK!
rest / full values saved

Guided calibration, every time you power on. The remote checks the trigger's rest position over 1.5 s of stable reading, without asking you anything. If the rest position has moved by more than 2% since last time (sensor, magnet, temperature), it also asks you to pull fully. It never blocks you: A skips the step and keeps the old value; no action for 6 s does the same. The motor can't spin during the sequence. Too short a travel is rejected (“TRAVEL TOO SHORT, magnet too far?”).

Power off

On the riding screen, **motor disarmed**, hold A: after 0.8 s the screen shows “Shutting down... keep A pressed”, and 1.2 s later the remote shows a big “See you soon”, one word per line, and goes to sleep. Letting go earlier cancels. **Power off is refused while the motor is armed**: you don't cut the control of a spinning motor. Before going to sleep, it asks the tracker to close the current session and freezes any running timer.

Auto sleep

After **20 minutes** with no action at all (button, trigger, touch), and only if the motor is disarmed, the remote switches itself off. Never while on the charger, never while the app is connected.

Charging

Plug in the charger: the “CHARGING” screen appears with the percentage and the voltage, and a blinking green dot. The motor is disarmed the moment you plug in, and riding is blocked. Bluetooth switches on by itself for the app (settings, update). Once it has been full for 2 minutes, the remote switches off; the charger can stay plugged in. Plugging in a remote that is off wakes it straight onto this screen.



Charging
% and voltage, Bluetooth on

Shutting down...
after holding A for 0.8 s

See you soon
2 s, then the remote goes to sleep

04 The riding screen

Two screens you can't mix up: **red with a padlock** while the motor is locked, **black with the throttle arc** as soon as it is armed. Touch is disabled on both screens: a wet hand or a wave can't trigger anything.



Disarmed
motor locked, A = arm (trigger at rest)



Armed, “speed” theme
speed in large, board battery underneath



Armed, “battery” theme
battery in large, speed underneath

Zone	Content
Top bar	**Radio signal** in 4 bars based on the real signal strength · **satellites** of the board's GPS (“8S”, cyan from 4 satellites) · **tracker dot** (option) · **time** (internal clock set by the app, corrected by the board's GPS), replaced by the timer while it runs · **remote battery** in %, green / orange / red
Centre	**Throttle arc** with a gradient that follows the trigger · **mode letter** L (green), M (yellow) or H (red) · large **GPS speed** with its unit · **board battery** in % (coloured by level) · **MAX**: top speed of the session, followed by the **estimated range** “25min 9km” as soon as the battery capacity is set (Wh remaining at standstill) · **L25** top right when a speed limit is set, in orange while the receiver is limiting
Bottom	**VOLT** pack voltage · **WATT** instant power · **TEMP** motor controller temperature, read live from the VESC. Replaced by **NO LINK** in red if the radio link drops.
Tracker dot	🟢 green: live session on sportstracklive.com · 🟠 orange: tracking started, waiting for GPS or 4G · ⚫ grey: tracking stopped · 🔴 red: account rejected · 🟡 yellow: no account · 🔴 **!** emergency alert in progress · none: no tracker
Banners	“● REC 00:42” under the top bar while a camera is recording · **arrow + distance** to the starting point, on the right under the top bar, as soon as you are more than 100 m away (section 10) · “START” for 1 s when the timer starts, then on stop the time in large for 3 s with the top speed, the distance, the number of falls and the longest flight · red bar along the bottom while you hold B towards emergency

The buttons on the riding screen

Gesture	Effect
A short	Arm (trigger must be at rest, otherwise a refusal beep) or disarm (throttle must be released: at full throttle the press is refused, to avoid disarming by accident)
B short	Next power mode: LOW → MED → HIGH → LOW, applied half a second after the press
B × 3 quick	Timer: start, or stop if it's running (section 11)
B hold 5 s	EMERGENCY page, red progress bar at the bottom of the screen; letting go earlier cancels
A hold 2 s	Power off, motor disarmed only
A + B together	Menu (150 ms of simultaneous press); same gesture to leave it

05 Motor safety

Protection	How it works
Disarmed start	Every time you power on and every time the remote wakes from sleep, the motor is locked. It takes a deliberate press on A, with the trigger at rest (less than 6% of travel).
Red screen	While the motor is locked, the screen is red with a big padlock: no possible doubt about the state, even out of the corner of your eye.
Link loss	Remote: disarms if there is no telemetry for 0.5 s. Receiver: motor to neutral if there is no command for 0.5 s. Fail-safe: it's silence that cuts the motor.
Auto disarm	Adjustable from 1 to 10 s (or OFF): with no throttle during that time, the motor locks itself. Ideal after a fall.
Disarm refused under throttle	Pressing A while the trigger is above 15% is ignored with a beep: release it first.
Power off refused when armed	You can't switch the remote off while the motor is armed.
Dead zone	A clean zero with hysteresis (7% to start, 4% to return): the motor doesn't twitch at rest.
Protected pairing	The receiver only accepts pairing during the 5 s after it is powered up, with the remote less than a metre away, and refuses any new remote while its own is connected. After that it only listens to its remote's address: a nearby eFoil can neither pick up nor send commands.
Charging	Plugging in the charger disarms immediately and blocks riding.
Touch	The touchscreen never responds on the riding screen, while charging, or with the motor armed.

06 Trigger, modes and riding settings

Power modes

Mode	Max power	Letter on screen
LOW	60%	L green
MED	80%	M yellow
HIGH	100%	H red

Switch with a press on B while riding, or set the default mode in the “Speed” menu. The mode is remembered.

Sensitivity (4 curves)

Level 0: nice and gentle (quadratic response, ideal for learning). Level 3: linear, direct response. Levels 1 and 2 are in between.

Startup boost (minimum thrust)

From 0 to 24% in steps of 3: the thrust applied as soon as you leave zero, so you take off cleanly without having to hunt for the trigger's starting point.

Learning and calibration

While riding, any reading beyond a known end stop becomes the new end stop, and is saved. The guided calibration every time you power on (section 3) protects against drift. The “Calibrate throttle” menu opens a full manual calibration in two guided steps (step 1 / 2 trigger released, step 2 / 2 trigger fully pulled, value in millivolts shown live, A confirms each step), for a first setup or after changing a magnet. B or the chevron cancels at any time. Green dot “OK!” then back to the menu; red dot “TRAVEL TOO SHORT” if the measured travel is under 150 mV (magnet too far?), and the old calibration is then kept.

Board battery

The percentage shown is calculated from the pack voltage and the number of cells set in the “Battery” menu (8 to 20 S), with a realistic Li-ion discharge curve. Visual alert below 44 V.

07 The menu

Press A and B together. The menu is a **carousel of coloured tiles**: the tile in the middle is the selected function, its neighbours are dimmed.



Menu, SportsTrackLive tile
2 / 21 counter at the top, status under the name, position dots



While swiping
the strip follows your finger, then snaps into place

By tilting (Gyro tile)

"Gyro" tile set to ON: three-step guided learning, then one gesture = **tilt clearly and come back** (threshold: 60 % of the learned tilt): right / left = next / previous, towards you = open, away from you = back. Never while riding (section 16).

Navigation

Gesture	Effect
Swipe left / right	Next / previous tile. The strip slides under your finger; beyond 40 px or with a quick flick it moves to the neighbour, otherwise it springs back.
B short	Next tile (animated slide)
Tap the tile or A	Open the function
Tap the left / right edge	Previous / next
Swipe right inside a page	Back (the smartphone "back" gesture)
A + B	Leave the menu, back to the riding screen

In simple pages: B changes the value (values loop around), A confirms and goes back. In switch pages: tapping the switch or B toggles it, the "‹" chevron or A goes back.

The 21 functions, in carousel order



Tile	Status shown	Role	Section
Pairing	GO	Link the remote to the board (guided procedure)	08
SportsTrackLive	ON / OFF / --	Live tracking of your session on the internet (tracker option)	09
Camera	REC / --	Choose the brand, then control the camera over Bluetooth	12
Bluetooth	OFF / visible / app	Link with the Lite Foil app and update	14
Timers	number	History of the last 10 timed runs: time, V.MAX, distance	11
Return	distance	Compass pointing to the starting point, remaining distance	10
Emergency	! / ON	Alert to your emergency contacts with your position (tracker option)	11
Throttle cal.	GO	Manual calibration of the trigger	06
Battery	12S...	Number of cells in the board's pack, 8 to 20 S	15
Speed	LOW / MED / HIGH	Default power mode	06
Limit	OFF · 12 to 40 km/h	GPS speed limit applied by the receiver	15
Capacity	-- · 10 to 100 Ah	Capacity of the board's battery, for the estimated range	15
Sensitivity	0 to 3	Trigger response curve	06
Boost	0 to 24 %	Minimum push when leaving zero	06
Units	km/h / mph	Speed and distance unit	15
Ping	ON / OFF	Shows the radio response time (diagnostics)	15
Gyro	ON / OFF / --	Tilt-controlled menus (guided learning when switched on)	16
Auto disarm	OFF, 1 to 10 s	Delay before automatic disarm without throttle	05
Theme	Black/Spd · Black/Bat	Speed or battery shown large on the riding screen	15
Language	FR / EN / ES / DE	Language of all screens	15
Distance	km or mi	Lifetime odometer of the remote (read only)	15

08 Pairing with the board

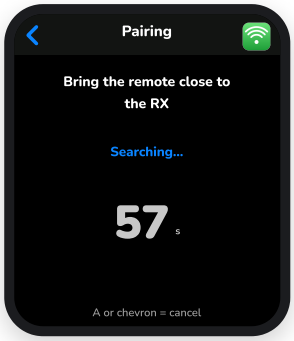
Do this once, when you first set things up or after changing the remote or the receiver. Pairing is stored on both sides and survives updates.

Procedure

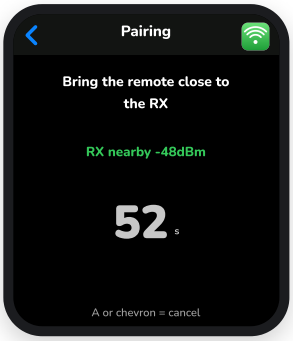
1. **Board off.** Turn on the remote, open the menu (A + B), “Pairing” tile, open it with A or by tapping the tile.
2. The page shows “Bring the remote close to the RX”, “Searching…” and a 60 s countdown. The remote is already broadcasting its request.
3. **Power up the board**, with the remote less than one metre from the receiver. The receiver only listens during its first 5 seconds.
4. “RX nearby −48 dBm” appears in green: the remote picks the receiver with the strongest signal, confirms it by name, and the receiver answers.
5. Green **PAIRING OK** badge for 1.5 s, then automatic return to the menu. Leave the menu (A + B): the red screen shows the link (signal, satellites, board battery).

NO RX after 60 s: the board was switched on too early or too late (5 s window), or too far away. Switch the board off, restart “Pairing”, then switch the board back on. A or the “‹” chevron cancel at any time.

Only one remote at a time. While the paired remote is connected, the receiver refuses any other request. To change remotes, switch the board back on with the old one turned off.



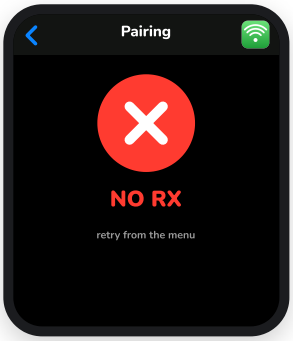
Searching
the board isn’t switched on yet



RX nearby
the receiver answers, strong signal



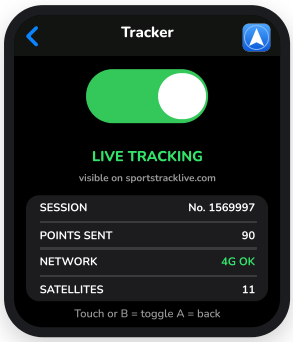
PAIRING OK
then back to the menu



NO RX
retry from the menu

09 SportsTrackLive: live tracking tracker option

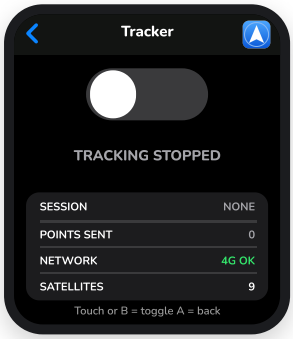
With the tracker installed in the board, your session is published **live on the internet**, with no phone on the water: someone on the beach or 500 km away can see where the rider is, their speed and their track, while they ride. The session then stays in their SportsTrackLive account (track, speed, max).



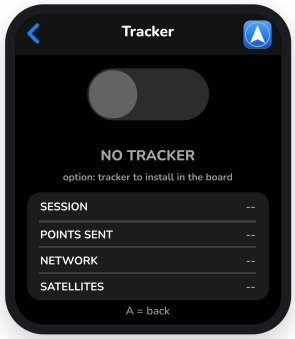
Live tracking
green switch, session no., points sent



Waiting for GPS / 4G
tracking started, session from the first point



Tracking stopped
tap the switch or B = start



No tracker
option to install, switch greyed out

How to use it

- **Once**: in the Lite Foil app, Tracker tab, enter the email and password of the SportsTrackLive account. They are stored in the tracker, not in the phone.
- **Every session**: menu, SportsTrackLive tile, flip the switch (tap or B). The tracker turns on its 4G, connects to the account and opens the session as soon as the first GPS point comes in: the switch turns green with the session number.
- **Stop**: same switch, or automatically when the remote switches off (immediately) or stays silent for 10 minutes.
- The **badge** in the top bar of the riding screen shows the status at all times (section 4).

Page states

LIVE TRACKING	visible on sportstracklive.com
WAITING FOR GPS / 4G	tracking started; the session starts at the GPS fix. The “Satellites” line stays orange while the fix is missing
TRACKING STOPPED	tracker present, nothing is published
ACCOUNT REJECTED	email or password rejected by the site: check them in the app
NO ACCOUNT	create the account in the app
NO TRACKER	board connected but without a tracker: option to install
NO LINK	board off or out of range

The tracker in a nutshell

Inside the board and powered by it: its own 5 Hz GPS, a 4G modem with an international SIM card, a secure connection to the site. With no network, the points wait and go out as soon as 4G comes back. After a power cut, it resumes a session less than 10 minutes old. A board sold without a tracker keeps exactly the same remote: the tracker can be added later with three wires.

10 Return to start

The board remembers **the spot where you first armed the motor** during the session. If you lose track of where the beach is, the “Return” tile shows a **compass**: the heading you are following, the direction of the starting point and how far away it is. There is no GPS in the remote: position and heading come from the board’s receiver, twice a second over the radio.



Riding
heading NE 47°, start 1.2 km away (green dot)



Stopped
heading frozen in orange: move to refresh it



No GPS
the board is still searching for satellites



Riding screen
arrow and distance from 100 m away from the start

Reading the compass

Item	Meaning
Graduated ring	Turns so the red **N** always points north; ticks every 2°, numbers every 30°, E S W in white
Index at the top	Your current direction. White: reliable heading · orange: frozen heading (board too slow)
Big numbers	Heading you are following, in degrees, with its name (N, NE, E…) underneath
Green dot	Direction of the starting point, linked to the centre by a dotted line. **Bring it to the top, under the index, and you are heading home.** Orange if the heading is frozen
Bottom line	HDG and its value · on the right, the distance to the start (metres, then kilometres beyond 1 km)

The heading comes from the GPS track, twice a second above 3 km/h. Between two positions and when you stop, the remote’s gyroscope turns the ring with you for up to two minutes; after that, the heading turns orange (frozen) until you get moving again.

Gestures

Gesture	Effect
B hold	Set the starting point **here** (if the board has a fix): “START HERE” shows in green for 1.5 s. Handy if you change spots or set off from a boat
A, chevron or swipe to the right	Back to the menu

States and messages

Message	Cause
start: at 1st arming	No starting point yet: it will be set at the first arming with a GPS fix, or with a hold on B
no GPS	The board has no position yet: wait under open sky
no link with the board	Radio cut off or board switched off
heading: receiver needs update	The board’s receiver runs firmware older than 14/10/2026: it does not send its position

The starting point is forgotten when you switch off. The return is a straight line: it is up to you to steer around obstacles; the dot gives you the direction again after every detour.

11 Emergency (SOS) and timer

Emergency alert

tracker option

From the riding screen, **hold** B **for 5 seconds**: a red bar grows along the bottom of the screen, then the EMERGENCY page opens. (You can also reach it from the “Emergency” tile in the menu.) On this page, **holding B for 3 seconds** fills the red bar and sends the alert: the tracker sends its GPS position to the emergency contacts of the SportsTrackLive account. The menu tile flashes and the red “!” badge shows in the top bar for as long as the alert is active. Live tracking is forced on and kept running, even if the remote goes silent.

To end it: on the EMERGENCY page, hold B for 3 s again. A leaves the page without triggering anything. Letting go of B before the bar is full cancels.



EMERGENCY page
B 3 s = send alert

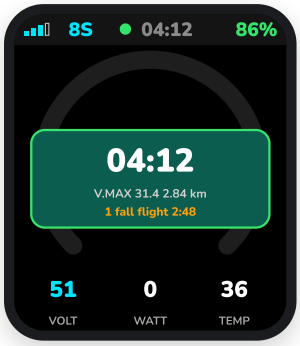


Alert active
red banner

Timer

While riding, **three quick presses on** B: a “START” banner for 1 s, then the elapsed time replaces the clock in the top bar (green, flashing colon). Three quick presses again: stop, the time is shown big for 3 s together with the **top speed** and the **distance** covered during the run, then it stays in grey for 30 s before the clock comes back. A single press keeps its usual job of changing the mode; two presses do nothing.

The timer keeps running if you disarm, is frozen when you switch off and shown again in grey the next time you switch on. The **last 10 timed runs** are stored in the “Timed runs” tile: date and time, time, V.MAX and distance in the unit you have set, **number of falls** (a jolt on the remote while the board was moving and then stops) and **longest flight** (continuous stretch above 8 km/h), the most recent one in green. One run per screen, big: swipe left / right, B or tilt to go from one to the next, A or the chevron to go back.



End of timer
time, V.MAX, distance, falls, flight (3 s)



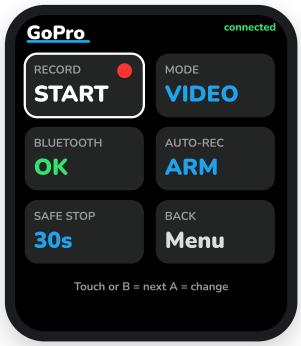
History
one run per screen in tiles, swipe to browse

12 Camera: GoPro and Insta360

The remote controls an action camera over Bluetooth: start and stop recording without letting go of the handle, or automate everything with motor arming or the timer.



Brand selection
GoPro HERO 9+, Insta360 X / ONE RS / Ace, Insta360 GO 3



Camera page (GoPro)
6 tiles, banner in the brand's colours

First connection

1. Put the camera in Bluetooth pairing mode (GoPro: Preferences › Connections › Connect Device › Remote; Insta360: Bluetooth turned on in the app).
2. Menu, "Camera" tile, choose the brand (tap the card, or B then A).
3. On the camera page, "BLUETOOTH" tile: the remote searches and connects. "connected" appears in green in the banner. The connection is remembered: next time, the camera reconnects by itself as soon as it is switched on.

The six tiles

RECORD	START / STOP. While recording, the tile turns red with the elapsed time, and "● REC" shows on the riding screen
MODE	VIDEO · PHOTO · TLAPSE (time-lapse)
BLUETOOTH	GO = connect the camera · OK = connected
AUTO-REC	OFF · ARM: recording starts when you arm the motor · TIMER: it follows the timer
SAFE STOP	Recording stops automatically 10 s, 30 s or 2 min after disarming, or OFF
BACK	Back to the menu

Navigation: tap a tile or B = next, A = change / run, hold A or swipe to the right = back. Cameras tested so far: GoPro HERO (Open GoPro protocol); Insta360 X and GO drivers included, validation in progress.

13 The Lite Foil app (iOS and Android)

One single app for the board, the tracker and the remote. It talks to the remote over Bluetooth and, through it, to the board. Everything you can set on the remote can also be set from your phone, on a bigger screen and with both hands free, and that's where you'll also find the timed runs, the logbook and the updates.

Install

Phone	How
Android	Google Play: search for **Lite Foil Tracker**. Or in Chrome, open **cedric167.github.io/litefoil-tracker** then “Add to Home screen”: same app, same icon.
iPhone / iPad	App Store: search for **Lite Foil Tracker**. Same tabs and same features as on Android.
App update	Automatic: every time you open it with internet, the new version loads by itself. Nothing to reinstall.
Account	No Lite Foil account. Only live tracking needs a SportsTrackLive account (section 09).
This manual	Always up to date in the app: **Remote** tab, **User manual** card, “Open” or “Download” (PDF).

Connect to the remote

1. Remote on: menu, **Bluetooth** tile, switch to ON, the screen shows “VISIBLE”. On the charger, it happens automatically.
2. In the app, **Remote** tab, **Connect** button, pick **LITEFOIL-REMOTE** in the phone's list.
3. The remote shows “APP CONNECTED” and receives the date and time from the phone. The trigger gauge and the mode move live in the app.

Range: a few metres. Bluetooth turns itself off after 5 minutes without a connection, and always while riding: the app is used on land, never on the water.

Settings from the phone

Settings card: all the menu settings (battery, speed, **GPS speed limit**, **battery capacity**, sensitivity, startup boost, units, auto disarm, theme, language, gyro, camera) in drop-down lists, then **Save to the remote**. The remote applies them right away and confirms with a beep. Gyro is only offered once the learning has been done on the remote (section 16). **Ring the remote** helps you find it in a bag.

The six tabs

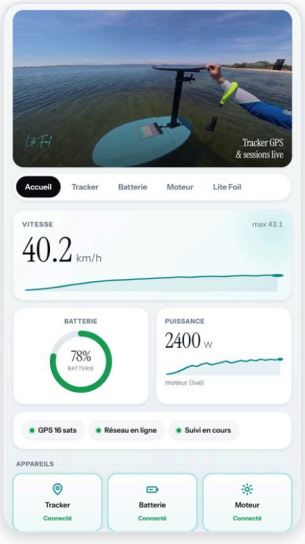
Tab	Contents
Home	Live summary: top speed, battery, power, GPS, network, tracking, and the status of the four devices (tracker, battery, VESC, remote). Community (WhatsApp, Facebook, Instagram), help and manual.
Tracker	SportsTrackLive account (login saved in the tracker), live status (GPS, 4G, session), start / stop tracking, advanced settings, **tracker firmware update** (section 14), tracker log.
Battery	Board pack: voltage, current, percentage, temperature, voltage of each cell, BMS protections.
VESC	VESC controller: voltage, motor and battery currents, power, temperature, RPM, faults.
Remote	Status (version, battery, link, **estimated range**, falls and flights of the session), settings, live trigger gauge and mode, timed runs, logbook, firmware update, user manual.
Lite Foil	litefoil.fr website: boards, accessories, user guide, contact, account, YouTube videos, access to SportsTrackLive.

Timed runs in the app

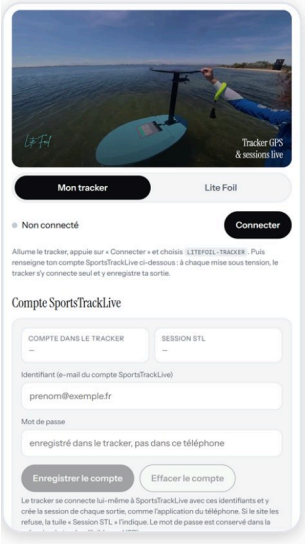
Timed runs card: the remote's last ten timed runs, dated, with time, top speed, distance, falls and longest flight. “Clear history” empties the list on the remote.

Logbook in the app

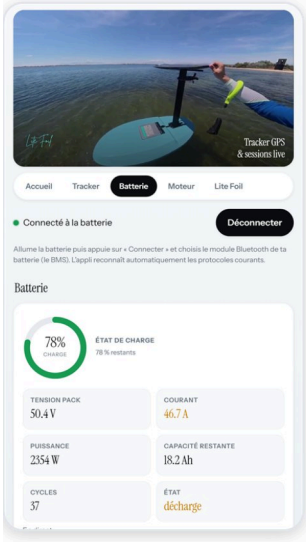
Logbook card: “List rides”, then “Open” a ride. Six tiles (duration, distance, top and average speed, motor time armed, start and end voltage), track on the map, three curves (speed, throttle, battery), **Export GPX** (Strava, Google Earth), “Delete this ride” or “Clear all”. Logbook details in section 14.



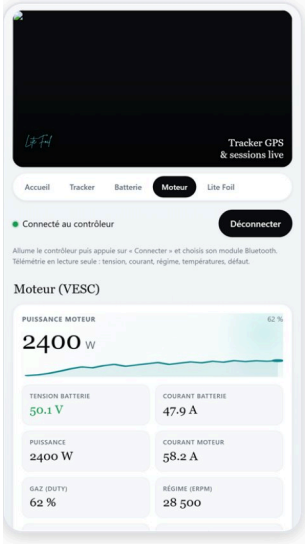
Home: live summary



Tracker: account and tracking



Battery: pack and cells



VESC: motor controller

14 Bluetooth, logbook and updates

Making the remote visible

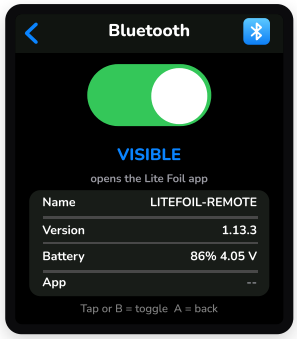
Menu, “Bluetooth” tile, flip the switch: the remote advertises itself as **LITEFOIL-REMOTE** for 5 minutes, then turns itself off if nobody connects. On the charger, it switches on by itself; while riding with the motor armed, it is always off. The page shows the name, the version, the battery and the app's signal.

Logbook

The remote records every ride in its memory, one line per second as long as the board is linked: time, GPS position, speed, throttle, pack voltage, remote battery, mode and armed state. 70 KB per hour, 3 MB kept (the oldest rides are erased automatically), one file per power-on. Transfer to the app over Bluetooth, ten seconds per hour of riding; nothing goes out to the internet.

Firmware update, step by step

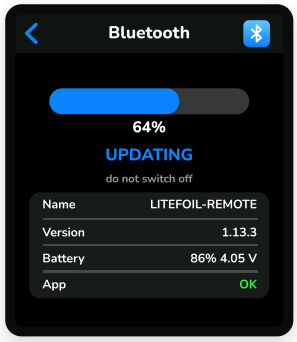
1. Remote on the charger (“CHARGING” screen, **Bluetooth already on**). Phone with internet.
2. App, **Remote** tab, “Connect”, pick LITEFOIL-REMOTE.
3. **Update** card: the installed version and the available version are shown, with what's new. Tap **“Update to x.y.z”**.
4. The remote's Bluetooth page shows a blue bar and “UPDATING, do not switch off”. Buttons and touch are ignored. About 2 minutes.
5. Automatic restart: “Hello”, logo, trigger calibration. The new version can be read on the Bluetooth page and in the app.



Visible
open the Lite Foil app



App connected
settings and live gauge



Updating
progress bar, nothing can be touched



Off
default state while riding

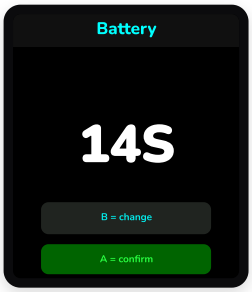
After the update	
Kept	Pairing with the board, trigger calibration, all settings, gyro learning, timed runs, odometer, logbook.
To redo	Nothing. If a version changes the format of the timed runs, the old history is simply reset; this is stated in what's new.
Interrupted during the transfer	The old firmware stays in place and the remote restarts on it: simply start the update again.
“No update”	The remote is already up to date, or the phone has no internet when the app checks.
Tracker	**Tracker** tab, Update card, board on, 3 to 6 min, automatic restart. First time (tracker from before October 2026): by USB cable on the receiver, then through the app.

If the connection doesn't work

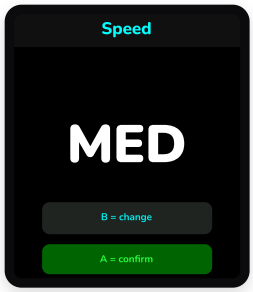
Symptom	Cause	What to do
LITEFOIL-REMOTE missing from the list	Remote's Bluetooth off (5 min elapsed, or motor armed)	Flip the Bluetooth switch again, or put the remote on the charger
The “Connect” button offers nothing	Browser without Bluetooth (Safari, Firefox) or phone's Bluetooth off	Use the app from Google Play or the App Store, or Chrome on Android; turn on the phone's Bluetooth
Connection lost after a few seconds	Too far away, or the app went into the background	Move closer, keep the app in the foreground during the transfer
The update stops halfway	Bluetooth dropped during the transfer	Nothing is broken: the old firmware restarts, start the update again
“Bluetooth” page: “--” or error	Bluetooth radio busy with the camera, or faulty	Flip the switch once (retry), otherwise switch the remote off and on again

15 Settings

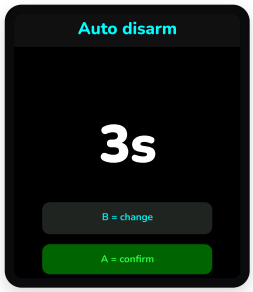
All setting pages have the same layout: the title, the value in large type, B = change (the values loop around), A = confirm and go back. Everything is saved immediately.



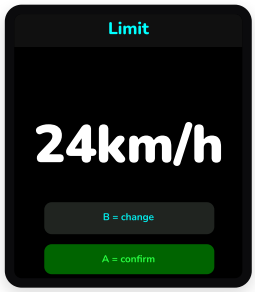
Battery
8 to 20 S



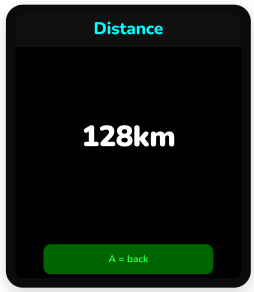
Speed
LOW · MED · HIGH



Auto disarm
OFF, 1 to 10 s



Limit
OFF · 12 to 40 km/h

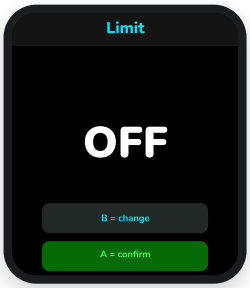


Distance
odometer, read-only

Setting	Values	Tip
Battery	8 S to 20 S	Number of cells in series in the board's pack. Essential for an accurate percentage. Lite Foil: 14 S.
Speed	LOW · MED · HIGH	Default power mode at power-on. Beginner: LOW.
Limit	OFF · 12 to 40 km/h (steps of 2)	GPS speed limit. The receiver smoothly lowers the throttle ceiling when the speed goes over the limit and raises it again below: the trigger stays proportional, never a cut-out. Accuracy 1 to 2 km/h (the GPS lags by one second). Designed for a child, a beginner or a rental; “L25” on the screen, orange when it is acting. Without a GPS fix, no limiting.
Capacity	-- · 10 to 100 Ah	Capacity of the board's battery. Enables the **estimated range**: the remote counts the Wh used since contact with the board, the average power (60 s) and the Wh per km, and shows “25min 9km” next to MAX while riding, the remaining Wh when stopped. The starting level comes from the resting voltage; board switched back on with another battery = counter reset to zero. An estimate, not a measurement: keep a margin.
Sensitivity	0 · 1 · 2 · 3	0 = very progressive, 3 = linear.
Startup boost	0 to 24 % (steps of 3)	Minimum thrust on the first millimetre of trigger. 6 to 9 % for a clean take-off.
Units	km/h · mph	Also changes the odometer and the timed run distances.
Ping	ON · OFF	Shows the radio response time on the riding screen. Diagnostics only.
Gyro	ON · OFF · --	Tilt menus (section 16). ON starts the three-step learning, A cancels it. “--”: sensor not detected. Can also be set from the app once the learning is done.
Auto disarm	OFF, 1 to 10 s	After a fall, the motor locks itself once the delay without throttle has passed. 3 to 5 s recommended.
Theme	Black/Spd · Black/Bat	Board speed or battery in large type in the centre of the riding screen.
Language	Français · English · Español · Deutsch	All screens and messages.
Distance	read-only	Distance travelled by the remote since it was made, added up from the GPS.

15 GPS speed limit and estimated range

Two settings added with firmware 1.15: a **speed limit** that the receiver enforces gently, and an **estimated range** that the remote works out from the battery capacity. Both are set in the menu (**Limit** and **Capacity** tiles, right after Speed) or from the app, Settings card of the Remote tab.



Limit: OFF
no limit (factory setting)



Limit: 24 km/h
B scrolls 12, 14 … 40, then OFF



Capacity: --
unknown, no range shown



Capacity: 30 Ah
10, 15 … 60, 70, 80, 100 Ah



While riding
orange "L24" = limiting in progress · "25min 9km"

GPS speed limit

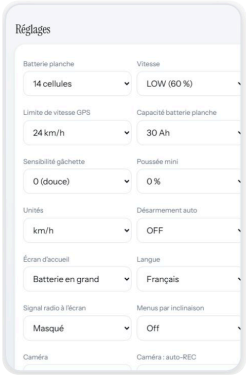
Set	**Limit** tile: B scrolls OFF, 12, 14 … 40 km/h (shown in mph if the unit is mph), A confirms. In the app: "GPS speed limit", from 8 to 40 km/h in 1 km/h steps.
How it works	The receiver compares the GPS speed with the limit 50 times a second. From 1 km/h below the limit, it lowers a **throttle ceiling**, all the faster the further you are over; below it, it raises it again slowly. The trigger stays proportional under the ceiling: you keep control, there is never a cut-out, and the ceiling never drops below 20 % so the board always has enough to keep going.
On screen	"**L24**" at the top right of the dial: **grey** = limit set, **orange** = the receiver is limiting. In the app, remote status: "limit 24 km/h (active)".
Accuracy	1 to 2 km/h above the limit: the GPS lags by about one second and braking is deliberately gentle so as not to jolt the rider.
Without GPS	No fix (at the start, overcast sky): no limiting possible, the ceiling stays at 100 % and "L24" stays grey.
Who it's for	Child: 12 to 15 km/h. Beginner: 18 to 20. Lending or rental: 25. For yourself: OFF.
Safety	The limit replaces neither LOW mode nor the safety features: releasing the trigger always cuts the motor, and auto disarm stays active.
Requirements	Updated receiver (October 2026). With an older receiver the limit is simply ignored. Older remotes remain compatible.

Estimated range

Set	**Capacity** tile: B scrolls --, 10, 15 … 60, 70, 80, 100 Ah; the capacity is written on the battery or its datasheet. In the app: "Board battery capacity". Leaving "--" turns the feature off.
Calculation	Pack energy = capacity (Ah) × cells × 3.6 V. On first contact with the board, trigger released, the resting voltage gives the starting level. Then the remote adds up the controller's watts as **Wh used**, works out the average power over the last 60 seconds and the **Wh per km** while flying.
On screen	Bottom line of the dial, next to MAX: "**25min 9km**" while riding (minutes at the average pace, km at the consumption rate), "**410Wh**" when stopped or as long as the average power stays below 100 W. In the app: "Estimated range".
Reset	Board off for more than 60 s (battery change) or remote switched back on: counter back to zero, new starting level from the resting voltage.
Accuracy	Good when you set off with a full battery. Battery half empty at the start: the initial level comes from the voltage curve, less accurate for the first few minutes. Wind, chop and weight change the Wh per km: it is an **estimate**, keep a 20 % margin to get back.
Tip	Set the capacity once and for all. When you get back, compare the % gauge with the range shown at the start; if they don't tell the same story, correct the capacity.



When stopped
grey "L24", "410Wh" remaining



App: both settings in the Settings card

16 Gyroscope: tilt menus, falls, flights, heading

The remote has a built-in 6-axis inertial unit (accelerometer + gyroscope, read 50 times a second, no magnetic compass). It is used for four things: navigating the menus by tilting the remote, counting falls, measuring flights, and turning the return compass between two GPS positions.

Turning it on: learning in three steps

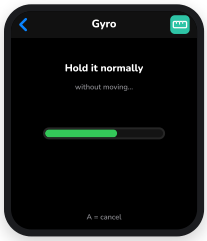
Menu, **Gyro** tile, B to switch to ON. The remote learns how *you* hold it: there is no "official" position.

1 · Hold it normally
without moving. As soon as your hand is steady, the green bar fills in 1 s.

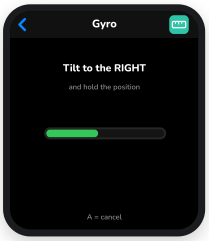
2 · Tilt to the RIGHT
and hold the position: a clear tilt (≈ 20°), held for 0.6 s. Beep.

3 · Tilt TOWARDS YOU
and hold: the same thing towards you. Beep, "ON", back to the menu.

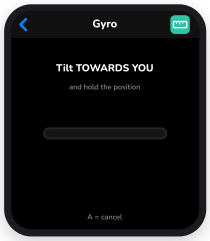
A cancels at any time (short beep, the tile stays OFF). With no gesture for 30 s, it gives up. Redo it if you switch hands. "--" on the tile: sensor not detected.



Step 1
still for 1 s



Step 2
to the right, 0.6 s



Step 3
towards you, 0.6 s

A gesture, how it works

A gesture = **tilt decisively, hold, then come back** to the starting position. The action fires on the way back, with a beep. The threshold is personal: **60 % of the tilt shown during learning** (at least 0.25 g, at most 0.55 g), axis by axis; tilting 25° while learning gives a threshold of about 15°. The acceleration is smoothed, so a jolt or a step won't trigger a gesture. The tilt must be held between 0.12 s (a tremor doesn't count) and 3 s (beyond that, the remote assumes your hand has changed position and starts again from there). You need 0.35 s between two gestures. After an update from a version older than 1.14.0, redo the learning to get the personal threshold. The rest position slowly follows your hand: holding the remote higher or lower triggers nothing, only decisive movements count. The rest position is picked up again at each screen change.

Never active on the riding screen, with the motor armed, on the charger, on the Emergency page or during a calibration. The Gyro setting itself can't be switched by gesture.

The four gestures, screen by screen

Screen	Right / left	Towards you	Away from you
Menu	next / previous tile	open the tile	leave the menu
Setting page	next / previous value	confirm	back
Tracker, Bluetooth	—	flip the switch	back
Timed runs	next / previous timer	—	back
Return	—	—	back
Camera	marker or next tile	choose / act	back

Falls

A **fall** is counted when the remote takes a knock (over 2.5 g) or drops in free fall (over 0.12 s) while the board was riding (at least 8 km/h in the previous 3 s) and then stops (under 4 km/h in the following 4 s). Short beep at each fall, at least 8 s between two falls. Counted per session (end-of-timer screen, app) and per timer (Timed runs page). A wave taken while riding without stopping doesn't count.

Flights

A **flight** is a continuous period at 8 km/h or more. It ends with a fall, or after 5 s below 4 km/h (deliberate stop). The Timed runs page shows the **longest flight** and the average time **between two falls** (timer time divided by the number of falls plus one).

Smoothed heading for the return

The heading comes from the board's GPS (twice a second, above 3 km/h). Between two positions and when stopped, the gyroscope turns the compass ring along with the remote, after learning its drift during 2 s of stillness. This gyro heading is accepted for 120 s after the last GPS position; after that the page shows "frozen" until the board picks up speed again.

Tips

Wide, decisive gestures, remote held normally. If gestures are missed or fire on their own: redo the learning, calmly, in your real riding position. When walking or on a boat, use the buttons instead.

17 Complete reference of the controls, screen by screen

A = left button, B = right button. Touch and tilt are never active while riding, with the motor armed or while charging. A + B together (150 ms): open the menu from the riding screen, and close it.

Screen	A short	A hold	B short	B other	Touch	Tilt
Off	—	3 s: switch on	—	—	—	— (plugging in the charger wakes it)
Startup calibration	skip the step, old value kept	same	—	—	—	—
Riding, DISARMED (red)	arm (trigger at rest, otherwise refusal beep)	0.8 s: “Shutting down...”, 2 s: switch off	mode LOW → MED → HIGH	× 3: timer start / stop · 5 s: Emergency page	inactive	inactive
Riding, ARMED	disarm (throttle released, otherwise refusal beep)	refused: you can’t switch off while armed	mode LOW → MED → HIGH	× 3: timer · 5 s: Emergency page	inactive	inactive
Charging	—	—	—	—	inactive	inactive (Bluetooth on only)
Menu (carousel)	open the middle tile	leave the menu	next tile	—	swipe: browse · tap the tile: open · left / right edges: previous / next	right / left: browse · towards you: open · away: leave
Setting page (Battery, Speed, Sensitivity...)	confirm and go back	go back without changing	next value (loops)	—	swipe right: back	right / left: value · towards you: confirm · away: back
Tracker, Bluetooth (switch)	back	back	flip the switch	—	switch: flip · ‹ chevron or swipe right: back	towards you: flip · away: back
Timed runs	back	back	next timer	—	swipe left / right: next / previous · chevron: back	right / left: next / previous · away: back
Return (compass)	back	back	—	hold: start point here (GPS needed)	chevron or swipe right: back	away: back
Emergency (SOS)	back	back	—	hold 3 s: send the alert, or cancel it if it is active	inactive	inactive
Pairing	cancel	cancel	—	—	chevron: cancel	—
Calibrate throttle (manual)	confirm the step	—	cancel	—	chevron: cancel	inactive
Gyro learning	cancel	cancel	—	—	—	the gesture is what is being learned
Camera, brand choice	choose the brand	back	next brand	—	tap a brand: choose · swipe left: next · right: back	right / left: brand · towards you: choose · away: back
Camera, tiles	act on the tile (REC, mode, connection, auto-REC, stop, back)	back	next tile	—	tap a tile: act · swipe right: back	right / left: tile · towards you: act · away: back

Sound signals

Sound	Meaning
Short beep	Action registered (tile, value, tilt gesture, timer).
Long beep	Refusal: arming with the trigger pulled, disarming under throttle, switching off with the motor armed.
Long beep + disarm	Radio link lost for more than 0.5 s: the motor is cut, the screen turns red.
Beep every 5 s	Board pack voltage below the warning threshold: head back.
Short beep while riding	Fall counted (section 16).

18 Messages, troubleshooting, maintenance

Message or symptom	Cause	What to do
Red DISARMED screen	Normal at switch-on: motor locked	Trigger at rest, press A
A press refused (long beep)	Trigger not at rest (arming) or throttle pulled (disarming)	Release the trigger completely, try again
NO LINK in red, "--"	No telemetry from the board for 0.5 s	Board on? Distance? Otherwise redo the pairing
Signal 1 bar	Remote far from the receiver or shielded by your body	Normal beyond a few dozen metres; the cut-out protects the rider
NO RX when pairing	Board switched on outside the 5 s window, or too far away	Start Pairing again, then switch on the board less than one metre away
TRAVEL TOO SHORT	Calibration: less than 150 mV between rest and full throttle	Check the trigger magnet, start "Calibrate throttle" again
hand on trigger?	At startup, the reading is far from the known rest position	Let go of the trigger; A skips the step
The remote won't switch on	Press too short, or battery flat	Hold A for 3 s; otherwise put it on charge, the CHARGING screen appears
It switches itself off	Automatic sleep after 20 min without any action, motor disarmed	Normal. Switch it back on with A for 3 s
Can't switch off	Motor armed	Disarm (A), then hold A for 2 s
ACCOUNT REFUSED (tracker)	SportsTrackLive e-mail or password rejected by the site	Check you can log in on sportstracklive.com, re-enter them in the app: the tracker retries straight away
WAITING GPS / 4G that lasts	The tracker has no GPS fix (Satellites line in orange) or no network	Board outdoors, clear sky; 1 to 2 min on first startup
NO TRACKER	Board connected, without a tracker	Option to be installed; everything else works
Bluetooth fault	The Bluetooth radio didn't start	B or tap the switch to try again; otherwise switch off / on again
Camera not connected	Camera off, out of range or not in pairing mode	Switch the camera on next to the remote, BLUETOOTH tile
Touch doesn't respond	Intentional on the riding screen, while charging and with the motor armed	Use A and B; touch works in the menu and its pages

Maintenance

- Rinse with fresh water after every session at sea, dry before charging.
- Charge with the supplied charger; don't leave the battery flat for several weeks.
- Don't leave the screen in the sun, switched off, in a closed car.
- Install the updates offered by the app: they bring fixes and features.

Warnings

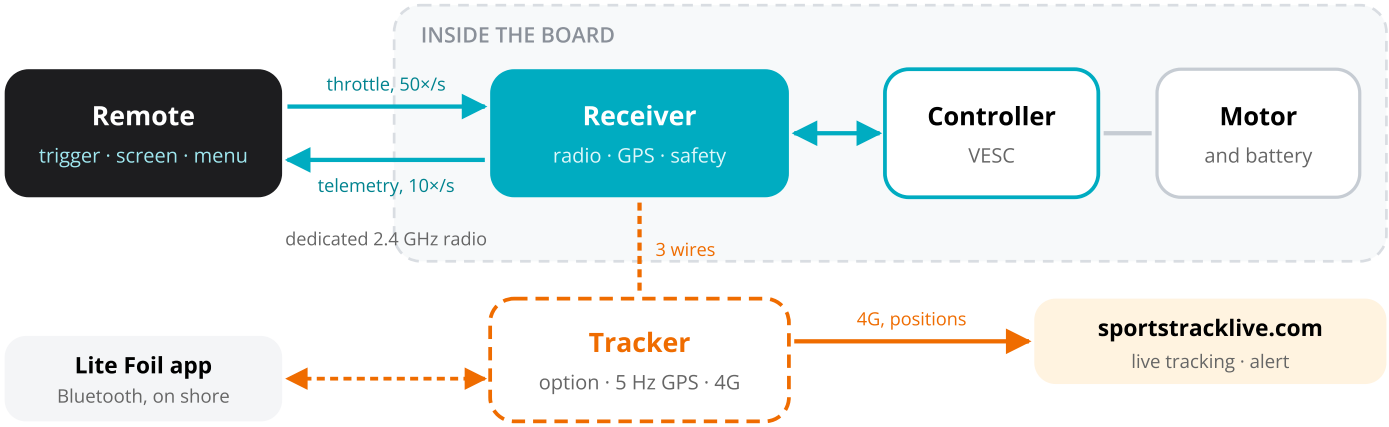
- Only ever arm the motor when you are ready, trigger at rest, board in the water and clear.
- Disarm before climbing back onto the board, going near the propeller or handing the remote to someone.
- Live tracking and the emergency alert depend on 4G coverage: they do not replace mandatory safety equipment.
- Wear the wrist strap: a lost remote cuts the motor within 0.5 s, but the board drifts.

19 The receiver: standard and tracker versions

The receiver is the electronic board housed in the board's compartment. It receives the remote's commands by radio, passes them to the motor controller (VESC) and sends back to the remote everything it displays: battery, power, temperature, GPS speed. It comes in **two versions**. The remote, the app and every control in this manual are the same for both.

Standard version
Radio control, safety features, board GPS, speed limit, return to start, motor telemetry.

Tracker version
Everything the standard version does, plus a GPS / 4G module: live tracking on the internet, emergency alert, rides kept in the SportsTrackLive account.



The two versions, feature by feature

Feature	Standard	With tracker	Details
Motor control by radio	✓	✓	Throttle received 50 times per second and passed to the controller
Safety (0.5 s cut-off, protected pairing)	✓	✓	Section 05; details on the next page
Telemetry: VOLT, WATT, TEMP, signal	✓	✓	Read live from the controller, sent back 10 times per second
Board GPS: speed, time, satellites	✓	✓	Large speed, MAX, corrected time, "8S" on the riding screen
GPS speed limit	✓	✓	Applied by the receiver (section 15)
Return to start	✓	✓	Position and heading sent twice per second (section 10)
Estimated range, timed runs, falls, flights	✓	✓	Calculated by the remote from the receiver's data
Live tracking on the internet	—	✓	On sportstracklive.com, with no phone on the water (section 09)
Emergency alert with position	—	✓	To the account's emergency contacts (section 11)
Rides kept in the online account	—	✓	Track, speed and max in the SportsTrackLive account
Tracker tab of the app	—	✓	Account, status, settings, update, over Bluetooth

Which version do I have? Remote menu, SportsTrackLive tile: "NO TRACKER" = standard version. On the riding screen, the tracker dot in the top bar only appears with the tracker version.

19 The receiver: features in detail

Standard version

Radio link

Dedicated 2.4 GHz radio at maximum transmit power. The receiver gets the trigger position 50 times per second and sends telemetry back 10 times per second. It only listens to its own remote's address: a nearby eFoil can neither listen in nor take control.

Fail-safe

With no command for 0.5 s, the receiver sets the motor to neutral; on its side the remote disarms. Silence is what cuts: remote lost, switched off or out of range, the motor stops.

Pairing

Accepted only during the 5 s after the board is powered on, with the remote less than a metre away. A single linked remote, stored and kept through updates; any other request is refused while it is connected (section 08).

Motor control and telemetry

The receiver passes the throttle to the motor controller (VESC) and reads from it, live, the pack voltage, the power and the controller temperature: these are the VOLT, WATT and TEMP at the bottom of the screen, replaced by NO LINK if the link drops.

Board GPS

Speed shown large and max speed of the ride, number of satellites, time that corrects the remote's clock, position and heading sent twice per second for the return compass.

GPS speed limit

The receiver compares the GPS speed with the limit set on the remote 50 times per second and smoothly lowers the throttle ceiling, never below 20 %, with no cut-off. Without a GPS fix, no limiting (section 15).

Starting point

Stored the first time you arm during the ride with a GPS fix, or reset by holding B on the Return page. Forgotten at power-off (section 10).

Speed limit and return to start need a receiver updated in October 2026. With an older receiver, the limit is ignored and the Return page shows "heading: update receiver".

Tracker version tracker option

The module

Inside the board, powered by it: its own 5 Hz GPS, 4G modem with an international SIM card, secure connection to the website, Bluetooth under the name **LITEFOIL-TRACKER** for the app.

Live tracking

Started from the remote's SportsTrackLive tile or from the app. The tracker turns on its 4G, logs into the account and opens the session at the first GPS point. Stopped by the same switch, when the remote is switched off, or after 10 minutes without it.

No network, nothing lost

Points wait and are sent as soon as 4G comes back. After a power cut, the tracker resumes the session if it is less than 10 minutes old.

Emergency alert

Hold B for 5 s on the riding screen, then B for 3 s on the EMERGENCY page: the GPS position goes to the account's emergency contacts, live tracking is forced on and kept on while the alert lasts (section 11).

SportsTrackLive account

E-mail and password entered once in the app, Tracker tab, checked and then written into the tracker, not the phone. "Erase account" removes them.

Status visible everywhere

Dot in the top bar (green: live, orange: waiting for GPS / 4G, grey: stopped, red: account refused, yellow: no account, !: alert). Tracker page: session number, points sent, network, satellites.

Advanced settings (app)

Interval between two points (1 to 60 s), delay between two uploads (5 to 3,600 s), SIM card APN, server address. A refused value keeps the previous one.

Update

Tracker tab, Update card, board switched on: 3 to 6 minutes, automatic restart. Tracker from before October 2026: the first time by USB cable on the receiver, then through the app.

Upgrading from standard to tracker. The tracker can be added to the board later with three wires. Nothing to change on the remote: pairing, settings, timed runs and logbook stay, and the SportsTrackLive tile goes from "NO TRACKER" to "TRACKING STOPPED". All that remains is to enter the account in the app.

Live tracking and the emergency alert depend on 4G coverage: they do not replace mandatory safety equipment.